

Mohith Munagala

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PROFESSIONAL SUMMARY

Cloud & DevOps Engineer with 5+ years of experience designing and managing **scalable cloud infrastructure across AWS and Azure**, with strong expertise in **Kubernetes (EKS, AKS), Docker, and microservices architecture**. Proven experience building and automating **CI/CD pipelines using GitHub Actions, Jenkins, GitLab CI/CD, and Azure DevOps**, enabling reliable and efficient deployments. Skilled in **Infrastructure as Code (Terraform, CloudFormation, Helm)** and implementing secure **cloud networking, IAM, RBAC, and Key Vault/KMS solutions**. Hands-on experience in developing **high-performance platforms supporting AI/ML workloads (LLMs, RAG) on GPU-enabled Kubernetes environments**. Strong background in **observability (Prometheus, Grafana, CloudWatch, Splunk)**, **automation (Python, Bash)**, and **DevSecOps practices**, delivering highly available, secure, and production-ready systems.

EDUCATION

Eastern Illinois University – Charleston, Illinois

Aug 2023 – May 2025

MS in Computer Science

Relevant Coursework: Cloud Computing, DevOps, Infrastructure as Code (IaC), CI/CD, Kubernetes, Scripting, Cloud Architecture, Security, AI Models and Automation Tools.

TECHNICAL SKILLS

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, VPC, IAM, KMS, CloudWatch, Route 53, API Gateway, SQS, EKS, AWS ECS Fargate, AWS ECR, AWS CDK, ALB), Microsoft Azure (AKS, App Services, Key Vault, Azure AD, ARM Templates)

DevOps & CI/CD: GitHub Actions, Jenkins, Azure DevOps, GitLab CI/CD, Argo CD (GitOps), AWS CodeDeploy

Infrastructure as Code (IaC): Terraform, AWS CloudFormation, Azure Bicep, AWS CDK, Helm

Containers & Orchestration: Docker, Kubernetes (EKS, AKS, On-Prem), OpenShift, Traefik Ingress Controller

Monitoring & Observability: Prometheus, Grafana, Amazon CloudWatch, SonarQube, Azure Monitor, Splunk, Datadog

Scripting & Automation: Python, Bash, Ansible, PowerShell, YAML, JSON

Security & Compliance: AWS IAM, Azure RBAC, KMS, Azure Key Vault, OAuth 2.0, S3 Bucket Policies, VPC Security Groups, HIPAA, GDPR

Networking: VPC, Public & Private Subnets, NAT Gateway, Route Tables, Internet Gateway, Security Groups, TCP/IP, CIDR, DNS, HTTPS, SSL/TLS, Nginx

Databases & Storage: PostgreSQL, Amazon RDS, MySQL, DynamoDB, Azure SQL, Elasticsearch, Neo4j

AI / ML & GPU Platforms: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Knowledge Graphs, Ollama, GPU Infrastructure, NVIDIA H200

EXPERIENCE



Amazon

Software Development Engineer – Cloud Engineer

Remote - USA

Apr 2026 – Present

- Designed and implemented scalable **AWS Cloud Infrastructure** using **AWS CDK, Amazon ECS Fargate, Application Load Balancer (ALB), Amazon RDS PostgreSQL, Amazon ECR, Amazon S3, and Amazon CloudWatch** across **DEV, TEST, UAT, and PROD** environments.
- Built and managed secure **AWS VPC Networking Architecture** with **Public/Private Subnets, NAT Gateways, Route Tables, Internet Gateway, Security Groups, and IAM-based Access Controls**.
- Developed reusable **Infrastructure-as-Code (IaC)** modules using **AWS CDK and TypeScript** for **Networking, ECS Services, Databases, Monitoring, Secrets Management, and Security Configurations**.
- Deployed and managed containerized microservices using **Docker** and **Amazon ECS Fargate**, supporting scalable **Frontend and Backend Application Deployments**.
- Created **ECS Task Definitions, ECS Services, Autoscaling Policies, ALB Target Groups, Health Checks, and Environment-Specific Deployment Configurations** for multiple microservices.
- Configured **Application Load Balancer (ALB)** routing, **SSL Integration, Path-Based Routing Rules, and Target Group Management** for secure and scalable traffic handling.

- Implemented **CI/CD Pipelines** using **GitHub Actions** for automated **Code Builds**, **Docker Image Creation**, **Amazon ECR Publishing**, **Amazon ECS Deployments**, and **AWS CDK Infrastructure Deployments**.
- Automated deployment workflows including **Docker Build Automation**, **ECR Image Publishing**, **ECS Rollout Validation**, **Deployment Verification**, and **Rollback Handling Processes**.
- Integrated **AWS Secrets Manager**, **AWS Systems Manager Parameter Store**, and **AWS KMS Encryption** for secure **Secrets Management** and **Environment Configuration Handling**.
- Built centralized **Monitoring and Observability Solutions** using **Amazon CloudWatch Dashboards**, **CloudWatch Alarms**, **ECS Metrics**, **ALB Metrics**, **RDS Monitoring**, and **Centralized Logging**.
- Collaborated with **Frontend**, **Backend**, and **Security Teams** to define **Infrastructure Standards**, **Deployment Strategies**, **Environment Configurations**, **Networking Requirements**, and **Operational Processes**.
- Developed local development environments using **Docker Compose**, **Mock Services**, **Environment-Based Configurations**, and **Local PostgreSQL Setups** to streamline developer onboarding and testing.
- Created **Deployment Runbooks**, **Operational Documentation**, **Disaster Recovery Procedures**, **Backup Strategies**, and **Support Guides** for **Production Support** and **Incident Management**.



Caterpillar
Cloud Engineer

Peoria, IL
Sep 2025 – Mar 2026

- Led design and deployment of **GPU-enabled Kubernetes (On-Prem) infrastructure** for enterprise AI platform, supporting **LLM inference workloads** with high availability and scalability.
- Architected and implemented **RAG (Retrieval-Augmented Generation) pipeline** integrating **LLMs**, **PostgreSQL**, and **Neo4j**, enabling context-aware AI applications.
- Deployed and managed **containerized applications using Docker and Kubernetes**, ensuring scalable **microservices architecture** with autoscaling and self-healing.
- Built and managed **Docker container images** for microservices and AI workloads, enabling consistent and portable deployments across **Kubernetes (on-prem clusters)**.
- Integrated **Docker image build and deployment workflows into CI/CD pipelines using GitHub Actions and Argo CD**, including image versioning and secure container registry management.
- Implemented **CI/CD pipelines using GitHub Actions and Argo CD (GitOps)** for automated build, test, and deployment across multiple environments.
- Designed and optimized **Kubernetes resource scheduling and GPU utilization**, improving application performance and infrastructure efficiency.
- Configured **secure cloud and on-prem authentication using Azure AD, RBAC, and Azure Key Vault**, ensuring compliance with enterprise security standards.
- Built and maintained **infrastructure automation and deployment workflows**, improving release reliability and reducing manual effort.
- Developed and maintained **Ansible playbooks** and roles for node provisioning, configuration management, and patch automation.
- Designed **ingress and networking architecture using Traefik, SSL/TLS, and load balancing**, ensuring secure service exposure.
- Implemented **distributed storage solutions using Longhorn** for persistent and highly available data storage across Kubernetes clusters.
- Established **monitoring and observability using Prometheus, Grafana, and Azure Monitor**, enabling proactive alerting and performance tracking.
- Built and managed **GitHub self-hosted runners** to support secure and scalable CI/CD pipelines for Kubernetes workloads.
- Conducted **performance tuning, capacity planning, and system optimization** for high-throughput AI applications.
- Performed **root cause analysis (RCA)** across **Kubernetes, networking, container runtime, and database layers** to ensure production stability.
- Improved **system reliability, scalability, and production performance** using DevOps and cloud automation best practices.
- Collaborated with cross-functional teams to implement **DevOps best practices, automation, and cloud-native solutions**.



British Airways
Cloud Engineer

New York, USA
June 2024 – Sep 2025

- Designed and deployed **AWS EKS-based Kubernetes infrastructure** using **Auto Scaling and ALB/NLB**, supporting high-traffic, production-grade flight booking applications.
- Provisioned and managed **cloud infrastructure using Terraform, AWS CloudFormation, and Helm**, enabling automated and scalable **Infrastructure as Code (IaC)** deployments.
- Built and secured **AWS VPC architecture (subnets, route tables, NAT Gateway, IAM, KMS)** ensuring compliance with enterprise security and governance standards.
- Built and optimized **CI/CD pipelines using GitHub Actions and Jenkins** for automated build, test, and deployment of **40+ microservices**.
- Implemented **Docker containerization and Kubernetes orchestration (EKS)** to deploy scalable, highly available **microservices architecture**.
- Designed and deployed **REST APIs using AWS API Gateway and AWS Lambda**, enabling serverless and event-driven application workflows.
- Developed **Python (Boto3) automation scripts** for **EKS scaling, S3 audits, and CloudWatch log management**, improving operational efficiency and reducing manual tasks.
- Implemented **blue/green and canary deployment strategies using Kubernetes and Helm**, enabling zero-downtime releases and improved deployment reliability.
- Configured **monitoring and observability using Prometheus, Grafana, AWS CloudWatch, and Splunk**, ensuring proactive alerting and system performance monitoring.
- Performed **performance testing and load testing using JMeter and Locust**, optimizing autoscaling and improving application performance under dynamic workloads.
- Integrated **SonarQube** into **CI/CD pipelines (Jenkins, GitHub Actions)** for **static code analysis**, enforcing **quality gates**, detecting **code smells, bugs, and security vulnerabilities**, and improving **code quality and secure deployments**.
- Designed and implemented **high availability and disaster recovery (DR) solutions across multi-AZ using Route 53 and CloudFormation failover mechanisms**.
- Built and maintained **secure SOAP APIs for IATA integrations**, ensuring compliance with airline data standards and external system interoperability.
- Collaborated with QA and Dev teams to implement **API testing using Postman, SoapUI, and automation scripts**, improving release quality and reducing defects.
- Participated in **Agile/Scrum processes**, including sprint planning, release coordination, and cross-team collaboration for continuous delivery.
- Applied **DevOps best practices, automation, and cloud-native architecture patterns** to improve system reliability, scalability, and deployment efficiency.



TCS
Cloud Engineer

Hyderabad, India
May 2021 – July 2023

- Provisioned and managed **AWS cloud infrastructure (EC2, VPC, S3, RDS, Lambda, IAM)** supporting scalable and secure financial applications.
- Automated **Infrastructure as Code (IaC)** using **Terraform and AWS CloudFormation**, enabling consistent deployments across **dev, UAT, and production environments**.
- Built and optimized **CI/CD pipelines using GitLab CI and AWS CodeDeploy** for automated build, test, and deployment of **containerized microservices**.
- Implemented **Docker containerization and Kubernetes orchestration (Amazon EKS)** to deploy scalable, highly available **microservices architecture**.
- Deployed and managed **microservices on Kubernetes (EKS)** using **Helm charts**, improving deployment automation and release consistency.
- Developed **serverless applications using AWS Lambda, API Gateway, and SQS** to support real-time payment processing systems.
- Created **Python and Bash automation scripts** for **EC2 lifecycle management, AMI backups, log cleanup, and CloudWatch monitoring integration**.

- Designed and enforced **AWS security best practices using IAM policies, KMS encryption, and S3 bucket policies**, ensuring compliance with enterprise standards.
- Configured **monitoring and observability using Prometheus, Grafana, and AWS CloudWatch**, enabling proactive alerting and performance monitoring.
- Performed **root cause analysis (RCA) and incident management using CloudTrail, VPC Flow Logs, and Splunk**, improving system reliability and uptime.
- Managed **API lifecycle (REST and SOAP APIs)** including versioning, authentication, and schema validation using cloud-native services.
- Maintained **runbooks, documentation, and automation repositories in Git**, supporting operational efficiency and team onboarding.
- Collaborated with cross-functional teams to implement **DevOps and DevSecOps practices**, improving deployment automation and system security.
- Participated in **Agile/Scrum processes**, delivering sprint-based cloud and DevOps solutions.
- Implemented and enforced **compliance and governance policies (HIPAA, GDPR)** in collaboration with InfoSec teams.

CERTIFICATIONS

- AWS Certified Solution Architect
- Microsoft Azure Fundamentals
- AWS Certified Cloud Practitioner
- Certified Kubernetes
- Certified Terraform